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Concise Report

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Palladium-Catalyzed Selective Syntheses of 2,3-Allenyl Amines *via* Double Functionalization Coupling of 2-Alkynyl-1,4-diol Dicarbonates

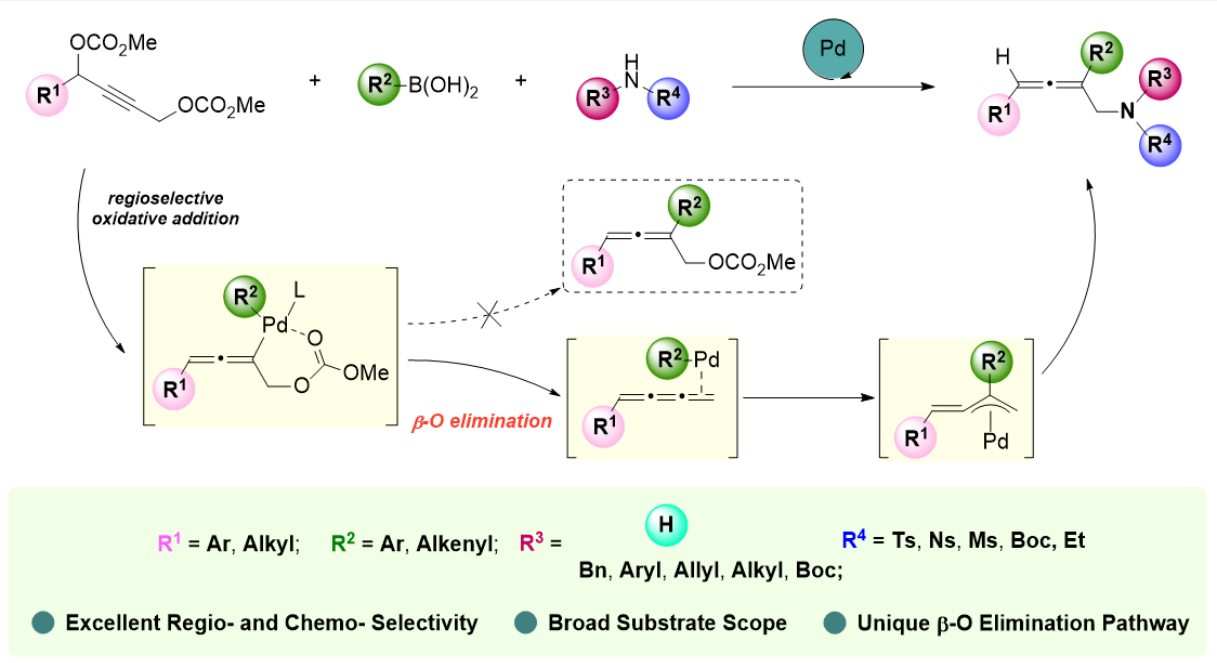
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Keywords

Pd catalysis | 2,3-Allenyl amine | Alk-2-yn-1,4-diol dicarbonate | Organoboronic acid | 1,2,3-Triene | Methylene-π-allyl palladium species | Regioselectivity | Chemoselectivity

Comprehensive Summary



2,3-Allenyl amines have shown wide applicability in biomedical and synthetic applications. Due to their enormous potential for applications, researchers have been dedicated to the development of methods for synthesizing 2,3-allenyl amines. Herein, a palladium-catalyzed three-component reaction of 2-alkynyl-1,4-diol dicarbonates, organoboronic acids, and nitrogen nucleophiles forming 2,3-allenyl amines with excellent regio- and chemo-selectivity has been developed. Substrate compatibility and synthetic applications have been demonstrated. Control experiments supported a mechanism involving 1,2,3-triene-Pd species and methylene-π-allyl palladium species.

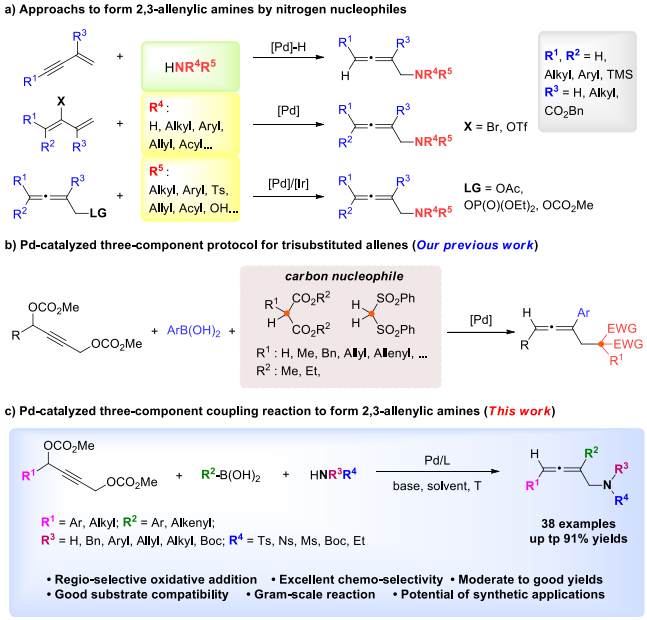
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Background and Originality Content

2,3-Allenyl amines have been attracting attention due to their application prospects.[1-2] Traditional approaches to form 2,3-allenyl amines applying nitrogen nucleophiles are two-component reactions between amines and enynes,[3] 2-(pseudo)halo-1,3-dienes,[4] and allenes[5] (Scheme 1a). Our group recently developed the Pd-catalyzed three-component coupling reaction of 2-alkynyl-1,4-diol dicarbonates, organoboronic acids, and malonate or bis(phenylsulfonyl)methane anions, which afforded trisubstituted allenes in excellent chemo- and regio-selectivity (Scheme 1b).[6] On the selection of nucleophiles, carbon nucleophiles such as malonates and bis(phenylsulfonyl)methane gave up to 91% yields, however, nitrogen nucleophiles are limited to Bn_2NH while other benzyl- and allyl- amines didn't work. Herein, we wish to report a Pd-catalyzed three-component coupling reaction yielding 2,3-allenyl amines from 2-alkynyl-1,4-diol dicarbonates, organoboronic acids, and nitrogen nucleophiles with an excellent selectivity (Scheme 1c).

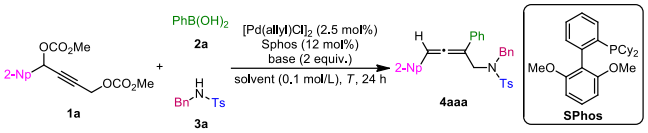
Scheme 1 Reactions forming 2,3-allenyl amines and three-component coupling *via* 2-alkynyl-1,4-diol dicarbonate



Results and Discussion

Initially, we tried to use 2-alkynyl-1,4-diol dicarbonate **1a**, phenylboronic acid **2a**, and *N*-benzyl-*p*-toluenesulfonamide **3a** as model substrates. When $[Pd(allyl)Cl]_2$, SPhos, and K_2CO_3 (Table 1, entry 1) were employed, we were delighted to observe a 70% yield of **4aaa**. With this result, various bases were examined (Table 1, entries 2—4) and Cs_2CO_3 turned out to be the best affording an 80% yield of **4aaa**. We then investigated the solvent effect: at first, we used MeCN, Et_2O, 1,4-dioxane, THF, and EtOAc to replace DCE (Table 1, entries 5—9), however, neither of them contributed a better result. To our delight, we finally found that when toluene was used as solvent, it could improve the yield of **4aaa** to 88% (Table 1, entry 10). Considering the possible impact of temperature on the reaction, the reaction was tried at 40 °C or 60 °C, however, lower yields of **4aaa** were observed (Table 1, entries 11 and 12). Further optimization was focused on the ratio of reactants: as the ratio of **1a** : **2a** : **3a** was changed from 1.0 : 1.0 : 1.5 to 1.2 : 1.2 : 1.0, however, there was no improvement of the yield of **4aaa** (Table 1, entry 13). Then we increased the concentration from 0.1 mol/L to 0.2 mol/L, and the yield of **4aaa** was improved to 90% (Table 1, entry 14). Next, we recalibrated the ratio of **1a** : **2a** : **3a** at a concentration of 0.2 mol/L, and to our delight, when the ratio of **1a** : **2a** : **3a** was changed to 1.0 : 1.1 : 1.5, the yield of **4aaa** was improved to 92% (Table 1, entry 15). Besides the ratio of the substrates, the loading of base was also optimized: applying 1.0 equivalent of Cs_2CO_3 only led to a slightly lower yield of **4aaa** (Table 1, entry 16), thus, we chose to employ this modified reaction conditions (Table 1, entry 16) as the standard.

Table 1 Optimization of reaction conditions[a]



entry	solvent	base	*T*/°C	NMR yield of **4aaa**[b]/%
1	DCE	K_2CO_3	50	70
2	DCE	K_3PO_4	50	72
3	DCE	*t*-BuONa	50	40
4	DCE	Cs_2CO_3	50	80
5	MeCN	Cs_2CO_3	50	—
6	Et_2O	Cs_2CO_3	50	29
7	1,4-dioxane	Cs_2CO_3	50	65
8	THF	Cs_2CO_3	50	71
9	EtOAc	Cs_2CO_3	50	79
10	toluene	Cs_2CO_3	50	88
11	toluene	Cs_2CO_3	40	85
12	toluene	Cs_2CO_3	60	81
13[c]	toluene	Cs_2CO_3	50	88
14[c,d]	toluene	Cs_2CO_3	50	90
15[d,e]	toluene	Cs_2CO_3	50	92
16[d,e,f]	**toluene**	**Cs_2CO_3**	**50**	**90**

Reaction conditions: [a] The reaction was carried out using **1a** (0.2 mmol), **2a** (0.2 mmol), **3a** (0.3 mmol), 2.5 mol% $[Pd(allyl)Cl]_2$, 12 mol% SPhos and base in 2 mL of DCE. [b] Determined by ^1H NMR analysis of the crude product using $(CHCl_2)_2$ as the internal standard. [c] 0.5 mmol scale reaction with **1a** : **2a** : **3a** = 1.2 : 1.2 : 1.0. [d] The concentration was 0.2 mol/L. [e] 0.4 mmol scale reaction with **1a** : **2a** : **3a** = 1.0 : 1.1 : 1.5. [f] 0.4 mmol Cs_2CO_3 (1.0 equiv.) was used.

With the optimized reaction conditions in hand, we began to investigate the substrate scope of three components, respectively (Scheme 2). Firstly, the scope of amines **3** was surveyed: with the electron-withdrawing group *p*-toluenesulfonyl (Ts) as R^4, R^3 may be benzyl (**3a**—**3c**), allyl (**3d**), alkyl (**3e**), 1*H*-indole-3-ethyl (**3f**), and 2-thienylmethyl (**3g**) affording the corresponding 2,3-allenyl amines **4** in 68%—89% yields. When *N*-Ts aniline **3h** was used as amine, the reaction afforded a 50% yield of 2,3-allenyl amine **4aah**. With electron-donating groups in the *para*-position of the benzene ring of the *N*-Ts aniline, the yields of 2,3-allenyl amines **4aai** and **4aaj** were improved to 60%—66%, while with electron-withdrawing group on the benzene ring, there was a significant decrease in the yield of corresponding 2,3-allenyl amine **4aak** (33%). For R^4, *p*-nitrobenzenesulfonyl (**3l**) and methylsulfonyl (**3m**) worked while groups such as *t*-butyloxy carbonyl (Boc), benzyloxycarbonyl (Cbz) and trifluoroacetyl failed to afford the corresponding allenes. Besides, bis(*tert*-butoxycarbonyl)amine (**3n**) showed good reactivity affording a 61% yield of target allene **4aan**. The structure of **4aan** was further confirmed by X-ray single crystal analysis. We also tried phthalimide, dimethylsulfoximine, or benzophenone imine as the amine **3**, separately. However, neither of them afforded the corresponding 2,3-allenyl amines **4**. Next, we investigated the reactivity of simple alkyl amines: Diethylamine **3o**

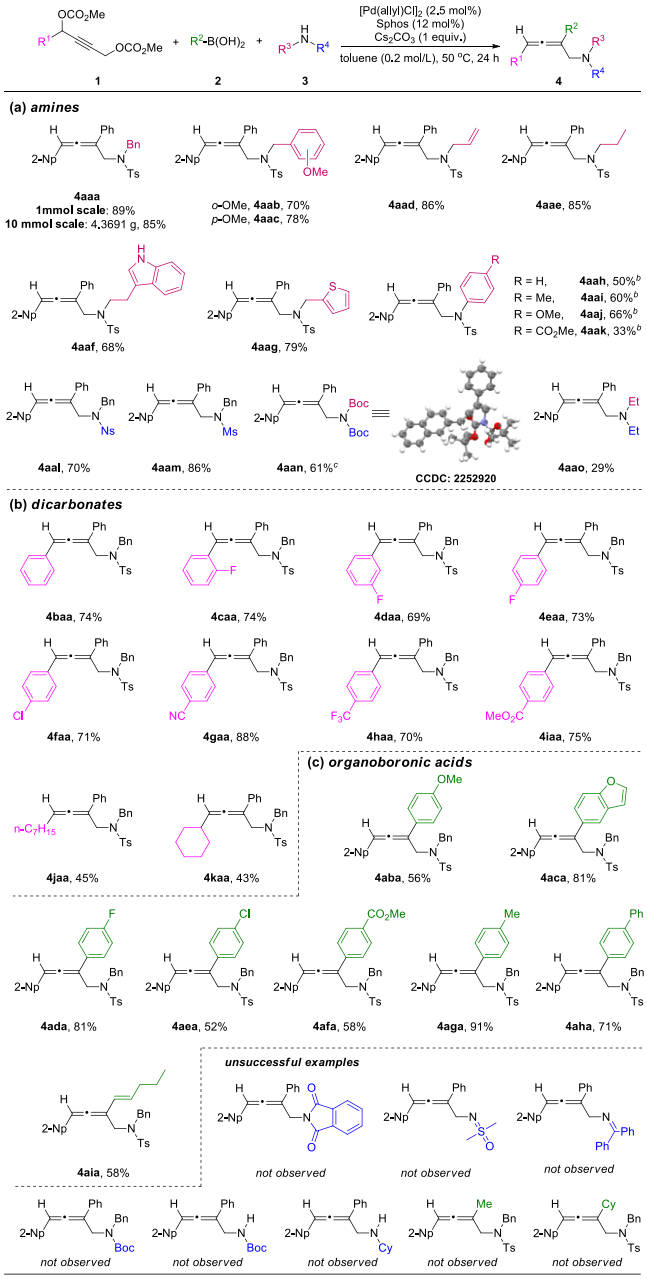
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showed moderate reactivity affording a 29% yield of 2,3-allenyl amine **4aao**. However, cyclohexylamine was not compatible with the reaction. The ten-mmol-scale reaction of **1a**, **2a**, and **3a** has also been verified, which generated 4.3691 g of **4aaa** in 85% yield.

Then, we investigated the substrate scope of 2-alkynyl-1,4-diol dicarbonates **1**. In addition to the naphthyl group, phenyl (**1b**) and the phenyl group substituted with different functional groups, such as halogen (**1c**—**1f**), cyano (**1g**), trifluoromethyl (**1h**), and methoxycarbonyl (**1i**) were also tolerated as R^1 with 69%—88% yields of **4**. In addition, we also examined the effect of fluorine atom substituted at different sites of the benzene ring (**1c**—**1e**), showing a slight difference in the yield of **4caa**—**4eaa** (69%—74%). For R^1, in addition to aromatic groups, alkyl groups, such as *n*-heptyl (**1j**) and cyclohexyl (**1k**), were also fine.

Scheme 2 Reactions forming 2,3-allenyl amines and three-component coupling *via* 2-alkynyl-1,4-diol dicarbonate[a]

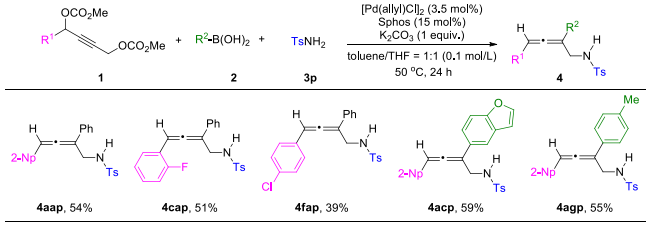


Reaction conditions: [a] The reaction was conducted using dicarbonate **1** (1.0 mmol), organoboronic acid **2** (1.1 mmol), and amine **3** (1.5 mmol) in 5 mL of toluene at 50 °C for 24 h. [b] Toluene/DCE = 1 : 1 (2.5 mL : 2.5 mL) was used. [c] Amine **3n** (2.0 equiv.).

The third part of the scope has been focused on organoboronic acids. The *para*-site of the phenyl group may bear methoxy (**2b**), halogen (**2d** and **2e**), methoxycarbonyl (**2f**), methyl (**2g**), and phenyl (**2h**) groups with 52%—91% yields of allenes (**4aba** and **4ada**—**4aha**). Benzofuran-5-ylboronic acid (**2c**) also showed good reactivity. Besides, alkenylboronic acid (**2i**) successfully worked affording a 58% yield of **4aia**. Alkyl boronic acid, such as methyl boronic acid and cyclohexyl boronic acid, was not compatible with the reaction.

To gain insights into the reactivity of primary amine **3p** in the reaction, we first chose 2-naphthyl (**1a**), 2-fluorophenyl (**1c**), and 4-chlorophenyl (**1f**) substituted dicarbonates to react with $PhB(OH)_2$ **2a** and $TsNH_2$ **3p**, and the target allenes **4aap**, **4cap**, and **4fap** were generated in 39%—54% yields (Scheme 3). Next, we tried to use benzofuran-5-ylboronic acid (**2c**) and 4-methyl phenylboronic acid (**2g**) in the reaction with 1-(2-naphthyl)but-2-yn-1,4-diyl dicarbonate **1a** and $TsNH_2$ **3p**, and the reactions afforded the target allenes **4acp** and **4agp** in 59% and 55% yields, respectively.

Scheme 3 Reaction scope of *p*-toluenesulfonamide (**3p**)[a]

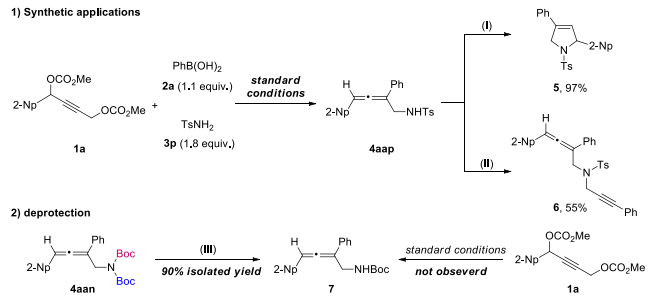


Reaction conditions: [a] The reaction was conducted using dicarbonate **1** (1.0 mmol), organoboronic acid **2** (1.1 mmol), and *p*-toluenesulfonamide **3p** (1.8 mmol) in 10 mL of solvent (toluene/THF = 1 : 1) at 50 °C for 24 h.

To demonstrate the synthetic potential of this reaction, we first prepared **4aap**. Then the cyclization reaction of 2,3-allenyl amine **4aap** generated *N*-tosyl-2,5-dihydro-1*H*-pyrrole **5** in a yield of 97%;[7] Mitsunobu reaction[8] of **4aap** with 3-phenyl-2-propyn-1-ol afforded *N*-(3-phenylprop-2-ynyl)-2,3-allenylamine **6** in 55% yield. On the other hand, we tried using $BocNH_2$ to form *N*-Boc-2,3-allenylamine **7** under the standard conditions. However, no target product was observed. So we tried to deprotect one of the two *t*-butyloxy carbonyls in **4aan** and found that 10 mol% $Sc(OTf)_3$ was able to remove one *t*-butyloxy carbonyl group in **4aan**,[9] and the corresponding deprotection product **7** was formed with 90% yield.

A series of control experiments were carried out to gain insights into the mechanism (Scheme 5). Firstly, when the reaction of 2-alkynyl-1,4-diol dicarbonate **1a** and phenylboronic acid **2a** was conducted under the standard condition in the absence of

Scheme 4 Synthetic applications and deprotection



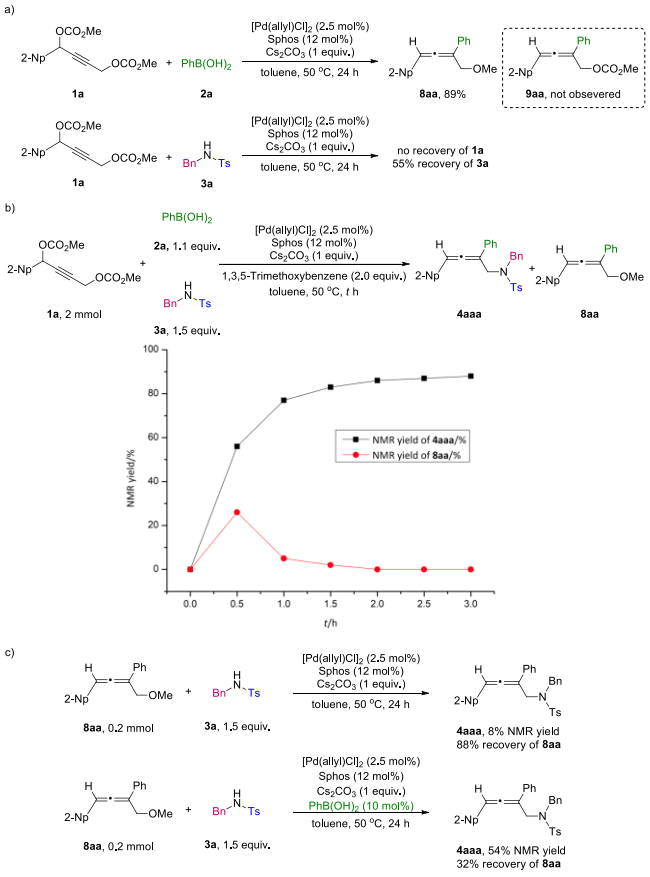
Reaction conditions: (I) **4aap** (0.2 mmol), $AgPF_6$ (10 mol%), DCM (0.1 mol/L), r.t., 17 h. (II) **4aap** (0.24 mmol), 3-phenyl-2-propyn-1-ol (0.2 mmol), PPh_3 (1.2 equiv.), DIAD (1.2 equiv.), THF (0.2 mol/L), 0 °C 15 min then r.t. 24 h. (III) **4aan** (0.1 mmol), $Sc(OTf)_3$ (10 mol%), THF (0.2 mol/L), r.t., 3 h.

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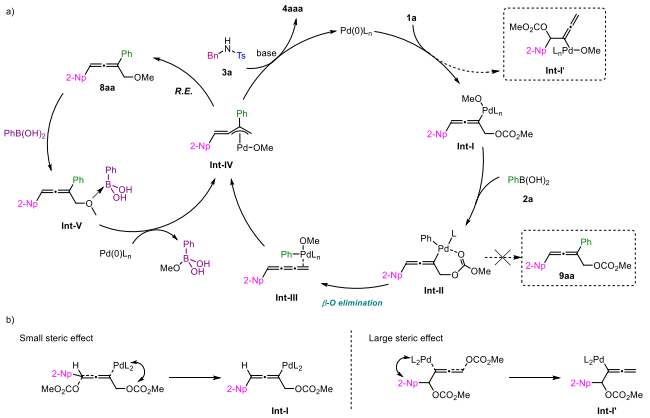
N-benzyl-*p*-toluenesulfonamide **3a** (Scheme 5a), 2,3-allenyl methyl ether **8aa** was observed with 89% yield while the formation of 2,3-allenyl carbonate **9aa** was not detected, which indicated that **9aa** was not the intermediate for this reaction. In the absence of phenylboronic acid **2a**, the reaction could not proceed with a 55% recovery of amine **3a** and no recovery of **1a** (Scheme 5a). Then we monitored the reaction of 2-alkynyl-1,4-diol dicarbonate **1a**, phenylboronic acid **2a**, and *N*-benzyl-*p*-toluenesulfonamide **3a** under the standard condition for 3 h (Scheme 5b). As shown in Scheme 5b, allenyl methyl ether **8aa** was *in situ* generated and converted to **4aaa** gradually. To examine whether 2,3-allenyl methyl ether **8aa** was the intermediate, we prepared **8aa** and tried its reaction with amine **3a** (Scheme 5c). However, surprisingly, an 8% NMR yield of 2,3-allenyl amine **4aaa** was formed, albeit most of **8aa** remained. Considering the absence of phenylboronic acid **2a** may cause the difference, we tried the reaction of 2,3-allenyl methyl ether **8aa** and *N*-benzyl-*p*-toluenesulfonamide **3a** with 10 mol% $PhB(OH)_2$ under the standard condition, and to our delight, a 54% NMR yield of 2,3-allenyl amine **4aaa** was afforded with a 32% recovery of **8aa**. This result indicated that phenylboronic acid played both the role of reactant and the activation of the C—O bond in intermediate 2,3-allenyl methyl ether **8aa**.

Scheme 5 Control experiments



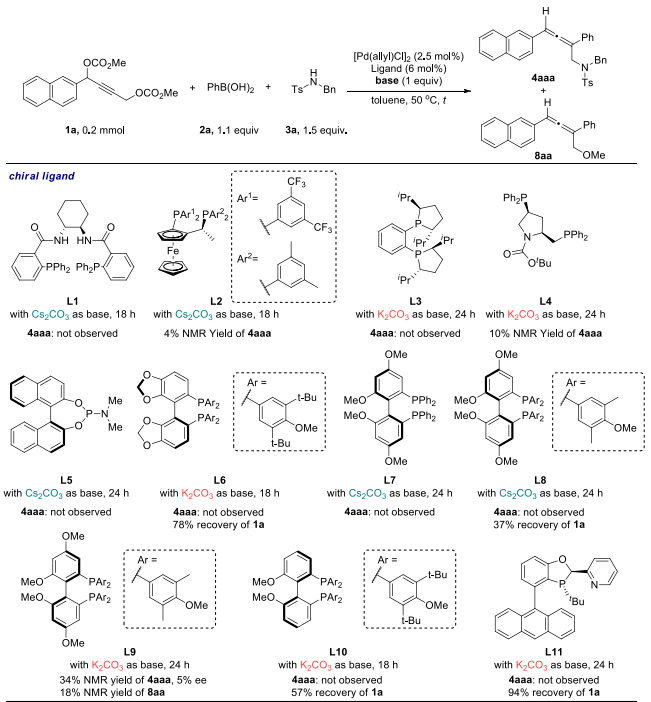
Based on these control experiments, we proposed a plausible mechanism as shown in Scheme 6a. Initially, the oxidative addition occurred between the catalytically active species $Pd(0)L_n$ and 2-alkynyl-1,4-diol dicarbonate **1a**. Allenyl palladium intermediate **Int-I** was generated exclusively, indicating that the oxidative addition occurred exclusively with the more sterically crowded C—O bond. Compared with its regioisomer **Int-I'**, the steric effect between palladium and substituent groups may cause this regioselectivity (Scheme 6b). After the transmetalation, allenyl phenyl palladium intermediate **Int-II** was afforded. Because of the stability of the six-membered cyclic coordination, this intermediate preferred β-O elimination to form 1,2,3-butatriene-coordinated palladium intermediate **Int-III** rather than reductive elimination, as supported by the absence of 2,3-allenyl carbonate **9aa** in the control experiment (Scheme 5a). Then, a regioselective insertion would lead to the formation of methylene-π-allyl palladium species **Int-IV**. **Int-IV** would be converted to 2,3-allenyl methyl ether **8aa** via reductive elimination. However, the C—O bond in **8aa** would be activated via coordination between $PhB(OH)_2$ and the oxygen atom of the OMe group in **Int-V**.[10] After the oxidative addition, methylene-π-allyl palladium species **Int-IV** was also regenerated. Finally, 2,3-allenyl amine **4aaa** was afforded via nucleophilic attack of the deprotonated *N*-benzyl-*p*-toluenesulfonamide **3a** to **Int-IV** and the Pd(0) species was regenerated to finish the catalytic cycle.

Scheme 6 Proposed mechanism



We have also tried many chiral ligands for this reaction (Scheme 7). However, most of them afforded a low yield and a poor enantioselectivity for the corresponding 2,3-allenyl amine **4aaa**.

Scheme 7 Screening of chiral ligands[a]



Reaction conditions: [a] The reaction was conducted using **1a** (0.2 mmol), **2a** (0.22 mmol), and **3a** (0.3 mmol) in 1 mL of toluene at 50 °C for 24 h.

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Conclusions

In conclusion, we have developed a palladium-catalyzed three-component coupling reaction to generate 2,3-allenyl amines with excellent regio- and chemo-selectivity. This reaction showed remarkable substrate tolerance of various 2-alkynyl-1,4-diol dicarbonate, organoboronic acid, and nitrogen nucleophiles. The synthetic potential of 2,3-allenyl amines has been demonstrated. Control experiments supported our unique proposed mechanism involving 1,2,3-triene-palladium species and methylene-π-allyl palladium species. Further studies on the regulation of enantioselectivity of this reaction are rather challenging and ongoing in our laboratory.

Experimental

Typical procedure: To a flame-dried Schlenk tube were added $[Pd(allyl)Cl]_2$ (0.025 mmol), Sphos (0.12 mmol), Cs_2CO_3 (1.0 mmol), **1** (1.0 mmol), **2** (1.1 mmol), and **3** (1.5 mmol). After the addition, the Schlenk tube was degassed and refilled with argon for three times at room temperature by vacuum, and toluene (5 mL) was added. The resulting mixture was stirred at 50 °C for 24 h as monitored by TLC. After cooling to room temperature, the resulting mixture was filtered through a short column of silica gel (3 cm) eluted with ethyl acetate (30 mL) and concentrated. The residue was purified by column chromatography on silica gel to afford the desired 2,3-allenyl amines **4**.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202400298.

Acknowledgement

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References

[1] For selected reviews on synthetic applications of 2,3-allenyl amines: (a) Alcaide, B.; Almendros, P. Novel cyclization reactions of aminoallenes. *Adv. Synth. Catal.* **2011**, *353*, 2561–2576; (b) Muñoz, M. P. Silver and platinum-catalysed addition of O-H and N-H bonds to allenes. *Chem. Soc. Rev.* **2014**, *43*, 3164–3183; (c) Ye, J.; Ma, S. Palladium-catalyzed cyclization reactions of allenes in the presence of unsaturated carbon–carbon bonds. *Acc. Chem. Res.* **2014**, *47*, 989–1000.

[2] For selected reports on applications of 2,3-allenyl amines: (a) Halliday, R. P.; Davis, C. S.; Heotis, J. P.; Pals, D. T.; Watson, E. J.; Bickerton, R. K. Allenic amines: a new class of nonhydrazine MAO inhibitors. *J. Pharm. Sci.* **1968**, *57*, 430–433; (b) Sahlberg, C.; Ross, S. B.; Fagervall, I.; Ask, A. L.; Claesson, A. Synthesis and monoamine oxidase inhibitory activities of α-allenic amines *in vivo* and *in vitro*. Different activities of two enantiomeric allenes. *J. Med. Chem.* **1983**, *26*, 1036–1042; (c) Castelhano, A. L.; Pliura, D. H.; Taylor, G. J.; Hsieh, K. C.; Krantz, A. Allenic suicide substrates. New inhibitors of vitamin B_6 linked decarboxylases. *J. Am. Chem. Soc.* **1984**, *106*, 2734–2735; (d) Sarhan, S.; Casara, P.; Knödgen, B.; Seiler, N. (4*S*)-4-amino-5,6-heptadienoic acid (MDL 72483): A potent anticonvulsant GABA-T inhibitor. *Neurochem. Res.* **1991**, *16*, 285–293; (e) Ohno, H.; Toda, A.; Miwa, Y.; Taga, T.; Osawa, E.; Yamaoka, Y.; Fujii, N.; Ibuka, T. First palladium-catalyzed aziridination reaction of amino allenes. *J. Org. Chem.* **1999**, *64*, 2992–2993; (f) Shu, W.; Yu, Q.; Jia, G.; Ma, S. Palladium-catalyzed three-component reaction of 2,3-allenyl amines, isocyanates, and organic halides: A diversified assembly of imidazolidinones. *Chem. Eur. J.* **2011**, *17*, 4720–4723; (g) Cheng, J.; Tang, X.; Yu, Y.; Ma, S. $FeCl_3$-catalyzed cyclization of α-sulfonamido-allenes with aldehydes—the substituent effect. *Chem. Commun.* **2012**, *48*, 12074–12076; (h) Li, S.; Ye, J.; Yuan, W.; Ma, S. Highly regioselective three-component palladium-catalyzed synthesis of 5-vinyloxazolidin-2-ones from 2,3-allenyl amines, organic iodides, and carbon dioxide. *Tetrahedron* **2013**, *69*, 10450–10456; (i) Luo, H.; Yang, Z.; Lin, W.; Zheng, Y.; Ma, S. A catalytic highly enantioselective allene approach to oxazolines. *Chem. Sci.* **2018**, *9*, 1964–1969; (j) Han, Y.; Qin, A.; Zhang, Q.; Zhang, X.; Qian, H.; Ma, S. Rhodium-catalyzed dynamic kinetic [4+2] cycloaddition of allene-1,3-dienes. *Angew. Chem. Int. Ed.* **2022**, *61*, e202211635.

[3] (a) Radhakrishnan, U.; Al-Masum, M.; Yamamoto, Y. Palladium catalyzed hydroamination of conjugated enynes. *Tetrahedron Lett.* **1998**, *39*, 1037–1040; (b) Brinkmann, C.; Barrett, A. G. M.; Hill, M. S.; Procopiou, P. A. Heavier alkaline earth catalysts for the intermolecular hydroamination of vinylarenes, dienes, and alkynes. *J. Am. Chem. Soc.* **2012**, *134*, 2193–2207; (c) Adamson, N. J.; Jeddi, H.; Malcolmson, S. J. Preparation of chiral allenes through Pd-catalyzed intermolecular hydroamination of conjugated enynes: Enantioselective synthesis enabled by catalyst design. *J. Am. Chem. Soc.* **2019**, *141*, 8574–8583; (d) Zhang, Y.; Yu, B.; Gao, B.; Zhang, T.; Huang, H. Triple-bond insertion triggers highly regioselective 1,4-aminomethylamination of 1,3-enynes with aminals enabled by Pd-catalyzed C-N bond activation, *Org. Lett.* **2019**, *21*, 535–539; (e) Zou, S.; Yu, B.; Huang, H. Palladium-catalyzed ring-closing aminoalkylative amination of unactivated aminoenynes. *Angew. Chem. Int. Ed.* **2023**, *62*, e202215325.

[4] (a) Ogasawara, M.; Ikeda, H.; Hayashi, T. π-Allylpalladium-Mediated Catalytic Synthesis of Functionalized Allenes. *Angew. Chem. Int. Ed.* **2000**, *39*, 1042–1044; (b) Ogasawara, M.; Ueyama, K.; Nagano, T.; Mizuhata, Y.; Hayashi, T. Palladium-catalyzed asymmetric synthesis of axially chiral (allenylmethyl)silanes and chirality transfer to stereogenic carbon centers in S_E' reactions. *Org. Lett.* **2003**, *5*, 217–219; (c) Ogasawara, M.; Ge, Y.; Uetake, K.; Takahashi, T. Vinyl ketones to allenes: Preparation of 1,3-dien-2-yl triflates and their application in Pd-catalyzed reactions with soft nucleophiles. *Org. Lett.* **2005**, *7*, 5697–5700; (d) Ogasawara, M.; Okada, A.; Nakajima, K.; Takahashi, T. Palladium-catalyzed synthesis of endocyclic allenes and their application in stereoselective [2 + 2] cycloaddition with ketenes. *Org. Lett.* **2009**, *11*, 177–180.

[5] (a) Imada, Y.; Nishida, M.; Kutsuwa, K.; Murahashi, S.-I.; Naota, T. Palladium-catalyzed asymmetric amination and imidation of 2,3-allenyl phosphates. *Org. Lett.* **2005**, *7*, 5837–5839; (b) Trost, B. M.; Fandrick, D. R.; Dinh, D. C. Dynamic kinetic asymmetric allylic alkylations of allenes. *J. Am. Chem. Soc.* **2005**, *127*, 14186–14187; (c) Imada, Y.; Nishida, M.; Naota, T. Sequential asymmetric homoallenylation of primary amines with a palladium catalyst. *Tetrahedron Lett.* **2008**, *49*, 4915–4917; (d) Boutier, A.; Kammerer-Pentier, C.; Krause, N.; Prestat, G.; Poli, G. Pd-catalyzed asymmetric synthesis of N-allenyl amides and their Au-catalyzed cycloisomerizative hydroalkylation: A new route toward enantioenriched pyrrolidones. *Chem. Eur. J.* **2012**, *18*, 3840–3844; (e) Wan, B.; Ma, S. Enantioselective decarboxylative amination: Synthesis of axially chiral allenyl amines. *Angew. Chem. Int. Ed.* **2013**, *52*, 441–445; (f) Li, Q.; Fu, C.; Ma, S. Palladium-catalyzed asymmetric amination of allenyl phosphates: Enantioselective synthesis of allenes with an additional unsaturated unit. *Angew. Chem. Int. Ed.* **2014**, *53*, 6511–6514; (g) Glatz, F.; Petrone, D. A.; Carreira, E. M. Ir-catalyzed enantioconvergent synthesis of diversely protected allenylic amines employing ammonia surrogates. *Angew. Chem. Int. Ed.* **2020**, *59*, 16404–16408; (h) Cui, Y.; Zhai, Y.; Xiao, J.; Li, C.; Zheng, W.-F.; Huang, C.; Wu, G.; Qin, A.; Lin, J.; Liu, Q.; Wang, H.; Wu, P.; Xu, H.; Zheng, Y.; Ma, S. Chirality memory of α-methylene-π-allyl iridium species. *Chem. Sci.* **2021**, *12*, 11831–11838; (i) Xiao, X.; Lu, Y.-J.; Tian, H.-Y.; Zhou, H.-J.; Li, J.-W.; Yao, Y.-P.; Ke, M.-L.; Chen, F.-E. Organocatalytic atroposelective *N*-alkylation: divergent synthesis of axially chiral sulfonamides and biaryl amino phenols. *Org. Chem. Front.* **2022**, *9*, 2830–2839; (j) Zha, T.; Rui, J.;

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Zhang, Z.; Zhang, D.; Yang, Z.; Yu, P.; Wang, Y.; Peng, F.; Shao, Z. Direct catalytic asymmetric and regiodivergent N1- and C3-allenylic alkylation of indoles. *Angew. Chem. Int. Ed.* **2023**, *62*, e202300844.
[6] Li, C.; Zhou, Z.; Ma, S. A Pd-catalyzed highly selective three-component protocol for trisubstituted allenes. *Chem. Sci.* **2023**, *14*, 7709–7715.
[7] Dieter, R. K.; Yu, H. Synthesis of 3-Pyrrolines, Annulated 3-Pyrrolines, and Pyrroles from α-Amino Allenes. *Org. Lett.* **2001**, *3*, 3855–3858.
[8] (a) Mitsunobu, O.; Yamada, M. Preparation of esters of carboxylic and phosphoric acid via quaternary phosphonium salts. *Bull. Chem. Soc. Jpn.* **1967**, *40*, 2380–2382; (b) Han, Y.; Zhao, Y.; Ma, S. Rhodium-catalyzed Pauson-Khand-Type cyclization of 1,5-allene-alkynes: A chirality transfer strategy for optically active bicyclic ketones. *Chem. Eur. J.* **2019**, *25*, 9529–9533.
[9] Huck, J.; Duru, C.; Roumestant, M. L.; Martinez, J. Parallel synthesis of new β^2-amino esters *via* conjugate nucleophilic additions. *Synthesis* **2003**, *14*, 2165–2168.
[10] Lyu, H.; Kevlishvili, I.; Yu, X.; Liu, P.; Dong, G. Boron insertion into alkyl ether bonds *via* zinc/nickel tandem catalysis. *Science* **2021**, *372*, 175–182.

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